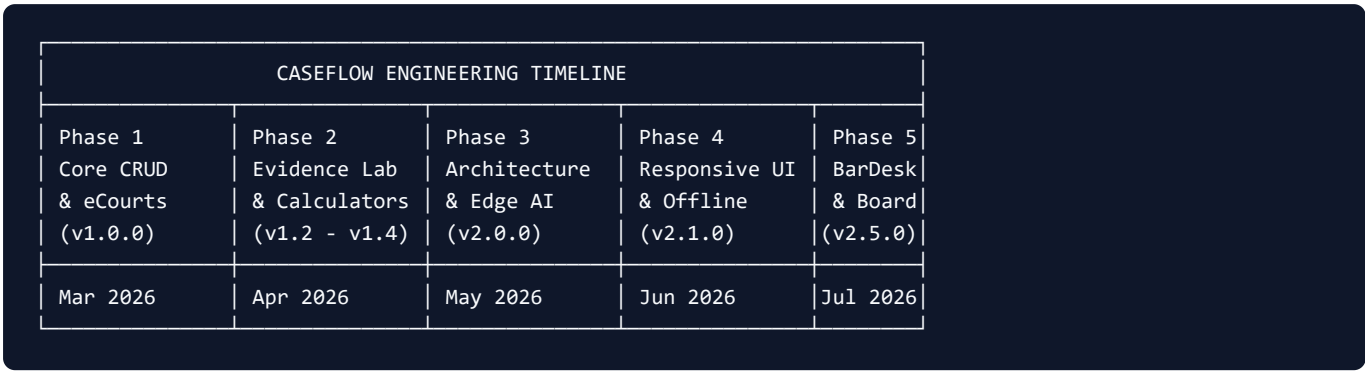
CaseFlow — Implementation Chronology & Order of Execution

Version 2.5 · Development Roadmap, Milestone Log & File Traceability
Confidential — For Project Managers & Lead Architects

1. Execution Overview & Timeline

CaseFlow was built iteratively following a **5-Phase Engineering Roadmap**, evolving from a basic case docket tracking utility into an enterprise legal tech platform.



2. Detailed Phase Breakdown & Order of Execution

Phase 1: Core Foundation & Case Management (`v1.0.0` · March 2026)

- **Goal:** Deliver basic case tracking, client records, and daily hearing reminders.
- **Key Implementation Steps:**

1. Configured Supabase Auth with PKCE and user role management (`profiles`). 2. Built `cases` , `clients` , `hearings` , `documents` , and `payments` tables with basic RLS policies. 3. Developed core Flutter mobile pages: `dashboard_screen.dart` , `cases_screen.dart` , `hearings_screen.dart` , `clients_screen.dart` . 4. Implemented eCourts scraper and CNR linking pipeline (`ecourt_parser_service.dart`). 5. Configured Firebase Cloud Messaging for basic push notifications (`fcm_service.dart`).

Phase 2: Evidence Lab, Commissioner Module & Legal Library (`v1.2.0-v1.4.0` · April 2026)

- **Goal:** Enable digital courtroom evidence gathering and offline legal reference tools.

- **Key Implementation Steps:**

1. Built Evidence Lab with live EXIF GPS overlay and local SHA-256 image hashing. 2. Implemented Local Commissioner module: field visit logs, vector canvas sketch tool (`commission_sketch_screen.dart`), and witness audio recordings. 3. Integrated Bare Acts library (BNS, BNSS, BSA, CPC, Constitution, DVC, NI Act, Motor Vehicles Act, Specific Relief Act). 4. Built 8 interactive calculators: Court Fee, Stamp Duty, Limitation Period, MACT, Maintenance, Partition, Interest, and Delay.

Phase 3: Architecture Refactoring, Security & Edge Proxies (`v2.0.0` · May 2026)

- **Goal:** Eliminate technical debt, secure third-party keys, and optimize state management.
- **Key Implementation Steps:**

1. **Monolith Refactoring:** Decomposed 986-line `SupabaseService` into 10 domain-specific repositories extending `BaseRepository` . 2. **Security Hardening:** Removed client-side Gemini and iLovePDF keys; routed queries through `gemini-proxy` and `pdf-proxy` Deno Edge Functions. 3. **Performance Optimization:** Added 5-minute TTL profile caching and `Future.wait()` parallel dashboard queries. 4. **CI/CD Pipeline:** Configured GitHub Actions workflows (`build-apk.yml` , `ci.yml`).

Phase 4: Responsive UI Engine & Offline-First Cache (`v2.1.0` · June 2026)

- **Goal:** Guarantee multi-device layout integrity and courtroom offline operation.
- **Key Implementation Steps:**

1. **ScreenUtil Migration:** Converted 110+ UI widgets to `flutter_screenutil` using `390×844dp` design baseline. 2. **Text Scaling Guard:** Clamped global text scale factor between `0.8x` and `1.15x` . 3. **SQLite Offline Cache:** Built `OfflineCacheService` using `sqflite` for offline read capability with automatic network sync.

Phase 5: Bar Association Module & Case Strategy Board (`v2.5.0` · July 2026)

- **Goal:** Enable bar community governance and collaborative Kanban task management.
- **Key Implementation Steps:**

1. **BarDesk Platform:** Built digital membership card generator with QR verification, election module, voter hash function, notices, and events. 2. **Case Strategy Board:** Created `strategy_tasks` , `task_comments` , `task_attachments` , `task_activity_log` tables with full RLS and Realtime subscriptions.

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3. File Traceability Matrix

Module	Core Logic & Repositories	UI Screens & Dialogs	Database / Edge Assets
Case Management	`case_repository.dart`	`cases_screen.dart` `case_detail_screen.dart`	`20260211102617_*.sq`
Court Diary	`hearing_repository.dart`	`hearings_screen.dart` `dashboard_court_diary.dart`	`send-hearing-remind`

Strategy Board	`strategy_repository.dart`	`case_strategy_tab.dart` `strategy_task_sheet.dart`	`20260729_strategy_b`
Evidence Lab	`evidence_repository.dart`	`evidence_lab_screen.dart` `commission_sketch_screen.dart`	`commission-evidence`
AI Assistant	`gemini_service.dart`	`ai_chat_screen.dart`	`gemini-proxy/index.`
Bar Association	`bar_association_repository.dart`	`bar_association_home_screen.dart` `bar_election_screen.dart`	`20260627_bar_associ`

4. Verification Checklist & Future Roadmap (v3.0.0)

- [x] **100% RLS Coverage:** All 38 database tables protected by Row Level Security policies.
- [x] **Zero Secret Leaks:** Gemini and iLovePDF API keys isolated in Deno secret storage.
- [x] **Responsive Adaptation:** Tested on 320dp to 430dp+ physical devices.
- [ ] **v3.0.0 Objective (Q4 2026):** High Court eFiling direct API integration and automated AI draft generation.

End of Implementation Chronology · CaseFlow Engineering Group